TITLE PROPOSAL DEFENSE

IntelliGate

An Edge-Artificial Intelligent License Plate Recognition and Vehicle Registration Portal for Automated Campus Security



PROPONENTS

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Background of the Study

Manual Verification Vulnerabilities

Guards primarily inspect school IDs visually, lacking capacity to verify driver credentials or vehicle registration paperwork.

Absence of Traffic & Access Logging

Boom barrier is operated via manual remote control without recording vehicle entry/exit timestamps or tracking identity.

The IntelliGate Solution

Automated Edge-AI access control combining real-time license plate OCR, vehicle color classification, and instant digital logging.

Statement of the Problem

General Problem: How to design and develop an automated multi-factor license plate recognition and access control system for campus security at Dr. Yanga's Colleges Elida Campus?

1. Localized Edge-AI System

Accurately recognize license plates and classify vehicle color for real-time dual verification.

2. Registration & Admin Portal

Develop database, mobile registration app, and web dashboard for entry/exit logging.

3. Gate Automation & Override

Automate boom barrier for verified vehicles while supporting manual RF remote guard override.

4. Performance Evaluation

Evaluate overall system reliability, speed, and accuracy compared to manual gate procedures.

Objectives of the Study

> 90%

Plate & Color Accuracy

Accurate plate reading and color classification (Objective 1).

≤ 2.0s

Processing Time

Average edge computing latency to open command (Objective 1).

100 %

Zero Breach Security

Zero invalid visitor uploads or unverified vehicles admitted (Obj 2 & 3).

Evaluation Framework

Measure performance, correct log rates, and ISO/IEC 25010 user feedback (Objective 4):

- Functionality & Reliability
- Usability & Security Ratings
- IT Expert vs. User Comparison

Significance of the Study

Administration
Vehicle database, timestamped entry/exit logs, and monitoring dashboard.

Security Guards
Less checking by sight; dashboard alerts guard to unverified vehicles.

Students & Employees
Register a vehicle once in app, with no repeated manual checking.

Visitors
Pre-registered visitors are recognized and admitted without delay.

Researchers
Applies computer vision, machine learning, and embedded systems.

Future Researchers & Schools
A reference for plate recognition with color verification.

Scope and Delimitation

Project Scope

- **Deployment Target:** Dr. Yanga's Colleges Elida Campus entrance.
- **Edge-AI Module:** YOLO plate detection & ML vehicle color classification.
- **Mobile Registration:** Student/Employee vehicle portal with OCR document reader.
- **Admin Web Interface:** Live vehicle logs, alerts, and visitor pre-registration.
- **Hardware Setup:** IP Camera, PC/Raspberry Pi 4, ESP32/Arduino, BLDC Motor, RF Override.

Delimitations

- **Single Lane Focus:** System designed for one entry/exit access point.
- **Vehicle Types:** Primary focus on standard cars and motorcycles.
- **Bench Test Prototype:** Boom barrier simulated via 24V BLDC motor setup.
- **Guard Fallback:** Unverified captures alert guards for manual RF remote override.

Theoretical Foundations & Literature

Deep Learning ALPR & OCR

YOLO single-pass object detection (Redmon et al.) with CRNN/EasyOCR character extraction (Shi et al.).

Edge Intelligence

Local execution on edge hardware cuts server latency and bandwidth (Satyanarayanan; Lin et al.).

Multi-Factor Verification

Pairing plate reading with color classification prevents plate-swapping exploits (Islas-Yañez et al.).

Research Gap Addressed

Combines edge dual-factor AI, document OCR mobile portal, live database whitelist, and manual RF override.

Conceptual Framework (IPO Model)

INPUT

- **Knowledge:** ALPR, OCR, ML color classifier, Edge-AI, App/Web dev.
- **Hardware:** IP Camera, PC/Raspberry Pi 4, ESP32, BLDC Motor, RF Module.
- **Software:** Python, OpenCV, YOLO, EasyOCR, MySQL/SQLite, Next/Flask.

PROCESS

- **Engineering Process:** Requirements, system architecture design.
- **Development:** AI training, document OCR, mobile & web app build.
- **Integration & Testing:** Hardware bench test & gate trials.

OUTPUT

IntelliGate System

An Edge-AI License Plate Recognition and Vehicle Registration Portal delivering automated security, visitor pre-registration, timestamped logs, and guard manual override.

Hypotheses of the Study (H01 - H06)

H01: Plate Recognition Accuracy
Does not achieve plate recognition accuracy meeting accepted level (>90%).

H02: Color Classification Accuracy
Does not achieve vehicle color accuracy meeting accepted level (>90%).

H03: Edge Processing Speed
No significant difference between edge latency and real-time threshold (≤2.0s).

H04: Document Verification
Does not correctly verify visitors from uploaded OR/CR and license documents.

H05: Gate Automation & Logging
Fails to correctly automate barrier action, dual verification, and override logging.

H06: Evaluator Ratings
No significant difference between IT experts and campus user evaluation ratings.

Hardware & Software Architecture

Hardware Stack

- **IP Camera / Webcam:** High-resolution visual capture at gate.
- **Edge Computing Node:** Local PC / Raspberry Pi 4 for AI execution.
- **Microcontroller:** ESP32 / Arduino Uno for serial barrier signals.
- **Barrier Mechanism:** 24V BLDC Motor / Relay module.
- **Manual Remote:** RF Receiver & Transmitter for guard override.

Software Stack

- **Computer Vision:** Python 3.10+, OpenCV video frame processing.
- **Detection & OCR:** YOLO plate detection, EasyOCR/Tesseract.
- **Color Classification:** Machine learning color classifier.
- **Database Engine:** Relational database (MySQL / SQLite).
- **User Portals:** Mobile App & Web Dashboard (Flask/Next.js).

Gate Operations Flowchart



Registration & Visitor Pre-Registration

Campus Member Registration

- **Document Upload:** Users upload Driver's License and vehicle OR/CR.
- **Text Extraction:** OCR extracts plate number, owner details, and vehicle color.
- **Database Entry:** Verified documents automatically populate whitelist.

Visitor Pre-Registration

- **Sponsor Submission:** Campus users pre-register visitors via mobile app.
- **Document Verification:** OCR verifies document text prior to gate arrival.
- **Admin Review:** Mismatched or unclear uploads refer to web admin review.

Sample Size & Testing Design

Testing Target	Sample Size	Sampling Rationale
Registered/Visitor Passes	**184 Passes**	Chow et al. (2008) 1–Proportion Formula
Unverified Gate Passes	**30 Passes**	20 Unregistered + 10 Color Mismatches
Visitor Document Submissions	**184 Submissions**	46 invalid document sets for fault testing
Volunteer Vehicles	**≥ 31 Vehicles**	Max 6 passes per vehicle limit (184 ÷ 6)
System Evaluators	**5 IT / 2–3 Guards**	Purposive Sampling for ISO 25010 survey

Testing Methodology

- ✓ Freeze code settings prior to gate trials.
- ✓ Bench testing table setup before installation.
- ✓ Tri-lighting runs (Morning, Afternoon, Night).

ISO/IEC 25010 Evaluation Questionnaire

Functionality
Reads plates & colors, opens barrier for authorized users, records exact timestamp logs.

Reliability
Consistent performance across lighting conditions; holds barrier closed when unverified.

Usability
Easy document uploads in mobile app, clear verification feedback, intuitive admin web dashboard.

Security
Blocks unregistered vehicles; protects personal data; maintains complete traceable audit logs.

Statistical Treatment of Hypotheses

Hypothesis	Data Source	Statistical Test	Decision Rule
H01 (Plate Accuracy)	184 gate passes	Exact Binomial Test	Reject if $p < 0.05$ (≥173 correct)
H02 (Color Accuracy)	184 gate passes	Exact Binomial Test	Reject if $p < 0.05$ (≥173 correct)
H03 (Processing Speed)	184 gate passes	One-sample t-test	Reject if $p < 0.05$ (≤2.0s mean)
H04 (OCR Decisions)	184 submissions	Exact Binomial Test	Reject if $p < 0.05$ (≥173 correct)
H05 (Barrier Action)	184 gate passes	Exact Binomial Test	Reject if $p < 0.05$ (≥173 correct)
H06 (Evaluator Ratings)	Survey Ratings	Independent t-test / Mann-Whitney U	Reject if $p < 0.05$

Likert Rating Scale

4.21 – 5.00: Strongly Agree
3.41 – 4.20: Agree
2.61 – 3.40: Neutral
1.81 – 2.60: Disagree
1.00 – 1.80: Strongly Disagree

Thank You!

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QUESTIONS & DEFENSE DISCUSSION